

The Emperor's New Password Manager: Security Analysis of Web-based Password Managers

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Zhiwei Li, Warren He, Devdatta Akhawe, and Dawn Song, *University of California, Berkeley*

We conduct a security analysis of five popular web-based password managers. Unlike “local” password managers, web-based password managers run in the browser. We identify four key security concerns for web-based password managers and, for each, identify representative vulnerabilities through our case studies. Our attacks are severe: in four out of the five password managers we studied, an attacker can learn a user’s credentials for arbitrary websites. We find vulnerabilities in diverse features like one-time passwords, bookmarklets, and shared passwords. The root-causes of the vulnerabilities are also diverse: ranging from logic and authorization mistakes to misunderstandings about the web security model, in addition to the typical vulnerabilities like CSRF and XSS. Our study suggests that it remains to be a challenge for the password managers to be secure. To guide future development of password managers, we provide guidance for password managers. Given the diversity of vulnerabilities we identified, we advocate a defense-in-depth approach to ensure security of password managers.

Zhiwei Li, University of California, Berkeley

Warren He, University of California, Berkeley

Devdatta Akhawe, University of California, Berkeley

Dawn Song, University of California, Berkeley

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